Where Is the Signal, and Where Is the Noise?

1. Candidate Information

Complete the fields that apply. Administrative information is not part of the technical score. Fields related to prior employment or public profiles may be left blank if not applicable.

Candidate detail	Response
Full name	Shivaraman T
Current location (City/State)	Sehore Madhya Pradesh
Willingness to work from Delhi / NCR	Both
Earliest possible start date	19/10/2026
Expected compensation	Negotiable, as per company standards
GitHub / GitLab / portfolio link (optional)	https://shivaraman-portfolio.vercel.app/
LinkedIn profile (optional)	https://www.linkedin.com/in/shivaramant
Resume attached: Yes / No	Yes

2. Build Experience: Your Work and Learning

Please list up to three products, features, tools, systems, or projects you built or contributed to. Coursework, lab experiments, hackathons, and team projects are welcome. A public repository is not required. Keep this short: one mini-profile per project is enough. This section supports the interview and is not part of the 90-minute technical exercise.

Project 1

ECG Signal Processing and Heart Rate Analysis — analyze ECG waveforms to explore heartbeat timing and estimate heart rate.

Worked on Python-based waveform analysis, filtering, peak detection, and frequency-domain exploration using NumPy, SciPy, Pandas, and Matplotlib. This was an assessment/project exercise; describe any team contributions accurately.

Python, NumPy, SciPy, Pandas, Matplotlib; repository listed in resume: https://github.com/SHIVARAMAN15/ECG-Signal-Processing-and-Heart-Rate-Analysis

Filtering can make a waveform look smoother without guaranteeing better timing. Peak detection and timing must be checked against the original waveform, and performance claims should not be made without validation.

Project 2

Smart Aeroponics System — monitor water/nutrient conditions and automate aeroponics control.

Worked on ESP32-based monitoring and control, sensor integration, and automated nutrient dosing/misting. State which elements were personally implemented and what teammates contributed.

ESP32; pH, TDS, temperature, and water-level sensing; ADC/GPIO/1-Wire; pumps and misting control. GitHub: https://github.com/SHIVARAMAN15/Smart-Aeroponics-System

Sensor calibration and end-to-end testing are important: verify each sensor and actuator independently before relying on integrated closed-loop control.

Project 3

Automatic Aerator System (AquaDox) — monitor dissolved oxygen and support automated aeration based on sensor readings.

Worked on a microcontroller-based sensor monitoring and threshold-driven aeration prototype; specify exact personal contribution and any team contributions.

Arduino Nano, dissolved-oxygen sensor, pH sensor, DS18B20 temperature sensor, HC-05 Bluetooth; GitHub: https://github.com/SHIVARAMAN15/Automatic-Aerator-System-AquaDox-

Sensor calibration, signal acquisition, and communication checks must be verified separately before relying on automated control decisions.

3. The Case: Five Short PPG Recordings

A PPG sensor detects changes in light related to the pulse at the skin. You are given five anonymous streams, S1–S5. Their conditions are deliberately not disclosed. Your role is not to diagnose a person or identify a vendor-specific failure, but to decide what can be measured reliably.

Supplied files: candidate_ppg_packets.json - Each recording contains nominally 6,000 samples at 200 samples/second. Every packet states its sequence number, first sample index, nominal first-sample time and sample values. Nominal sample time = sample index / 200. Amplitudes are in arbitrary units.

Case 1 — Artifacts, Filtering and Measurement Integrity

Problem: Measurements can contain real pulse information and unwanted effects from motion, electronics, sampling, or handling. The goal is to reduce the unwanted part without damaging the information you actually need. Your job is to decide what "unwanted" means.

Q1A. Observe and form hypotheses

Plot all five unmodified 30-second streams. Pick two that look meaningfully different. For each recording, describe what you notice in the waveform and when it happens. Describe exactly what changes across time, frequency and other domains of your choice. Write two competing explanations for at least one issue, a prediction that distinguishes them, and the smallest test that would change your mind.

Explain how the observed issues could affect pulse counting, the timing of a pulse peak, the area the pulse, the morphology of a pulse and any other downstream activities that would be done typically on these signals. Based on your observations, would you Accept, Review, or Reject the affected data for pulse measurement? Briefly explain why. Your decision is provisional; some parts of a recording may still be useful.

Choose one unusual observation and suggest two possible explanations. What simple check would help you distinguish between them?

Analysis and observations (based on the supplied dataset): S1 and S2 have relatively smooth, repeating pulse-like waveforms. S3 and S4 show more rapid fluctuations superimposed on recognizable pulse patterns. S5 has a pulse-like waveform but contains a one-second missing interval from 10 to 11 seconds. These are signal observations only; they do not establish a physical cause.

Packet integrity: S1–S4 each contain 60 packets and 6,000 valid samples, with no missing sample indices detected. S5 contains 58 packets; sequence numbers 20 and 21 are missing, corresponding to 200 samples (1 second) between 10 and 11 seconds. The received packet values were finite; the missing interval is absent data, not evidence of 200 corrupted received values.

Comparison of S1 and S3: S1 has first-difference standard deviation about 0.02718, while S3 is about 0.18905 (approximately 6.95 times higher). After diagnostic low-pass filtering below 5 Hz, their low-frequency components had standard deviations about 0.453619 and 0.453589 and correlation about 0.999995. This supports a similar slow pulse pattern plus additional faster variation in S3, but does not identify the source.

Competing explanations: (1) electrical/acquisition interference adds a high-frequency component; (2) movement/contact variation or another signal-dependent effect creates rapid fluctuations. A small discriminating test is to repeat a controlled stationary recording and a deliberate-movement recording while comparing raw ADC samples, stored samples, and spectra. A persistent narrowband component during stationary acquisition would support, but not alone prove, an electrical source.

Downstream effects and provisional decisions: S1—Accept provisionally for exploratory pulse measurements, with normal validation; S2—Accept provisionally if pulse detection and quality checks pass; S3—Review because rapid variation may cause false peaks or alter peak timing, height, width, and area; S4—Review because rapid fluctuations need source investigation and measurement validation; S5—Review and mark the 10–11 s interval invalid for pulse measurements that depend on continuous data. Reject only a measurement or segment that fails predefined quality criteria; do not assume every portion of a recording is unusable.

Unusual observation and check: S5's two consecutive missing packets could be caused by transmission/buffer loss or by storage/parsing omission. Compare sender and receiver sequence counters and inspect the original packet log to locate where the packets disappeared.

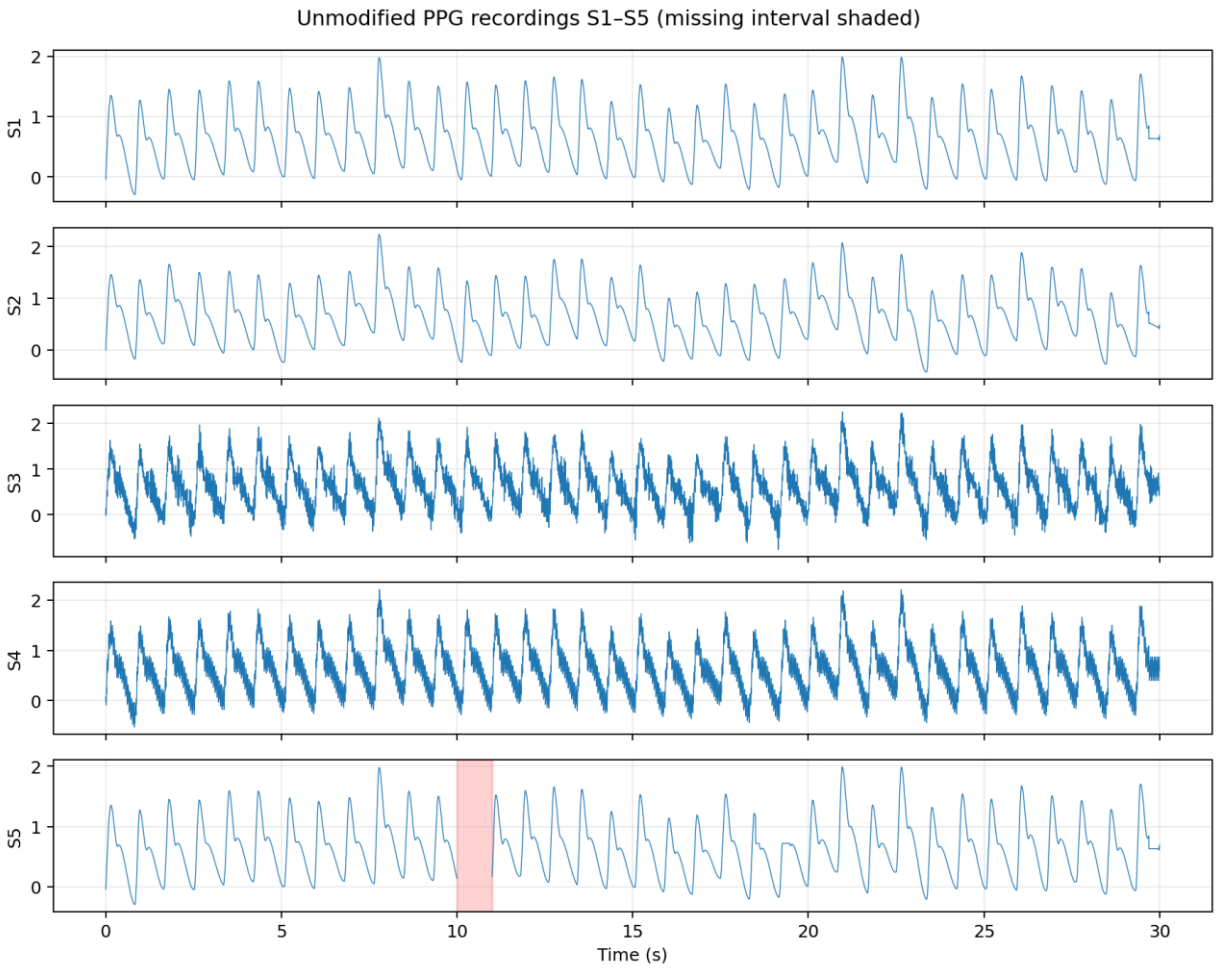


Figure 1. Unmodified recordings S1–S5. The shaded interval indicates the known S5 gap from 10 to 11 seconds.

Q1B. Choose, test and challenge one mitigation

Select one of the five recordings where the pulse waveform contains some visible disturbance, but individual pulses can still be recognized. Choose one processing method to improve the fidelity of the signals.

Before applying it, explain

1. What changes and observations do you expect?
2. What information should remain unchanged?
3. What result would tell you that your approach is not working?

Implement your selected method. Plot the same interval before and after. Identify the same pulse in both waveforms and compare: the pulse timing, height, weight and other morphological descriptors. Use graphical, programmatical calculations to support your comparison. If you cannot reliably measure something, explain with justifications.

Finally, based on your observations, explain

1. Did your method actually improve the measurement, how did you verify it?
2. What remains uncertain, and what additional check would help you make a better decision?

Selected mitigation: fourth-order 8 Hz low-pass Butterworth filter applied to S3 using forward-backward filtering (sosfiltfilt). This is an exploratory offline method, not a clinically validated setting.

Expectation and invariants: reduce rapid fluctuations while retaining pulse order, approximate timing, and broad pulse morphology. On the full 30 s record, raw and filtered standard deviations were 0.488225 and 0.457763; first-difference standard deviations were 0.189055 and 0.026712. The first-difference standard deviation fell by 85.9%.

Peak check: with the same exploratory detector settings (minimum distance 0.45 s, prominence 0.40), the raw signal produced 37 peaks and the filtered signal 36. A one-to-one nearest-neighbour comparison with a 150 ms matching tolerance matched 36 events; median absolute timing difference was 15.0 ms and maximum was 45.0 ms. One raw peak remained unmatched near 29.97 s.

Morphology and failure criteria: filtering can change peak height, half-prominence width, and pulse area, even when events remain aligned. A result would be unacceptable if it suppresses a genuine pulse, introduces false events, shifts timing beyond a tolerance specified for the intended application, or materially distorts morphology. The unmatched event at 29.97 s is near the recording boundary and could be noise, a real feature, or affected by edge behavior; it is flagged for review, not automatically classified as false.

Assessment: the filter reduced rapid variation and retained most detected events in this experiment, but improvement in measurement accuracy is not yet proven. Next checks: inspect the unmatched event; compare pulse height, width, and fixed-window area using consistent event pairing; repeat on other disturbed recordings and across the entire dataset; and define acceptance tolerances before evaluating performance.

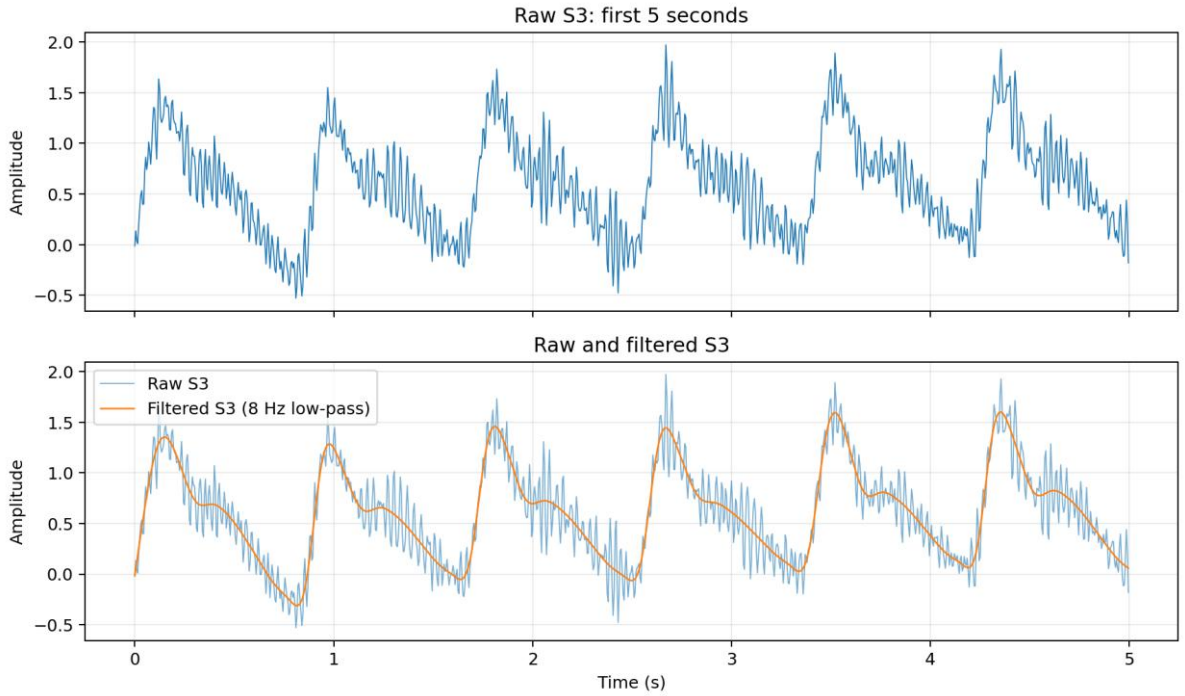


Figure 2. S3 raw and filtered comparison. Filtering smooths the waveform; this alone is not proof of improved physiological accuracy.

Q1C. A spectrum and sampling thought experiment

Imagine a PPG signal (not necessarily the ones supplied) containing useful components up to 30 Hz and a periodic disturbance at 10 Hz, with motion components between 0.5Hz – 2 Hz. It was sampled at 200 Hz with no analog prefilter. Sketch the expected observed spectrum from 0 to 100 Hz and label where the disturbance components appear. Which components overlap potentially useful signal? Explain, in simple words, what a filter could remove, what it might also damage, and what must instead be prevented or measured earlier in the acquisition chain. What filters would you prioritize and why? A labeled sketch and explanation are enough.

Sampling and filtering explanation: fs = 200 Hz, so the Nyquist frequency is 100 Hz. Useful content extends to 30 Hz; motion-related components are at 0.5–2 Hz; the periodic disturbance is at 10 Hz. Both the 10 Hz disturbance and motion-related components overlap potentially useful signal frequencies, so a simple frequency-only filter cannot remove them without risking useful information. A narrow 10 Hz notch may help if the disturbance is stable, but may also remove genuine signal content around 10 Hz.

There is no analog prefilter in the thought experiment, so unwanted analog frequencies above 100 Hz could alias into the sampled 0–100 Hz range. Digital filtering after sampling cannot reliably undo aliasing that has already occurred. Priorities: improve sensor contact/mechanical stability; investigate the source of the 10 Hz component; add an analog anti-alias filter before the ADC, designed to preserve the required 0–30 Hz band and attenuate out-of-band energy; then validate any digital notch/filter against pulse timing and morphology.

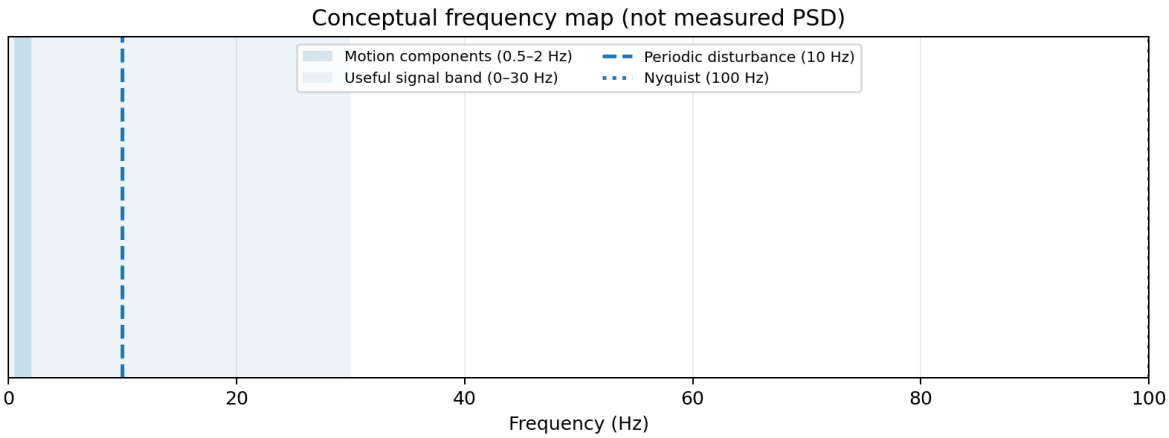


Figure 3. Conceptual frequency map; it labels specified bands and is not a measured power spectrum.

Case 2 — Trace the Fault Through the Whole System (20 minutes)

A colleague says: “The waveform looks wrong, so the signal-processing algorithm must be broken.” Another says: “The hardware is noisy, so software should fix it.” Neither explanation is established. Consider this simplified chain: light/tissue/contact → optical sensor and amplifier → analog filtering → ADC/sampling → firmware timestamps/buffering → data packets/storage → software preprocessing → event measurements and quality decision.

Choose TWO suspicious observations from Case 1. For each, give (a) two plausible fault locations at different stages; (b) what evidence you would request to distinguish them; (c) one hardware/firmware prevention or diagnostic action and one processing-level mitigation or safe failure rule; and (d) what test would verify the correction. You may recommend any reasonable improvement, but prioritize the FIRST three checks rather than writing a full system redesign.

Observation 1 — S5 packet gap: S5 has 58/60 packets, missing sequence numbers 20 and 21, and 200 missing sample indices between 10 and 11 seconds. Possible locations at different stages are (a) firmware/transmission/buffer loss and (b) storage or preprocessing omission. Evidence: compare sender/receiver sequence counters, buffer-overflow/drop logs, original packet records, and reconstructed sample indices. Prevention/diagnostic: sequence numbers, sample indices, counters, and robust buffering. Safe software behavior: preserve the gap as missing, flag affected measurements, and never silently present interpolated values as measured samples. Verify by deliberately dropping packets and checking detection, timestamps, quality flags, and downstream handling.

Observation 2 — S3 high-frequency variation: S3 has much greater rapid variation than S1 while the low-frequency components are highly similar after diagnostic filtering. Possible locations are (a) optical contact/motion at the sensor interface and (b) electrical interference or acquisition behavior in the analog front end/ADC. Evidence: controlled stationary-versus-motion recordings, raw ADC samples versus transmitted/stored values, repeated PSDs, and power/acquisition diagnostics. Prevention/diagnostic: improve mounting/contact and inspect grounding/power and analog bandwidth. Software mitigation/safe failure: monitor high-frequency energy and signal quality, apply a validated filter only where justified, and retain a review flag if confidence remains low. Verify on repeated controlled recordings and compare pulse timing, height, width, area, and event counts.

Prioritized first three checks: (1) verify S5 packet loss against sender/receiver logs and the original packet file; (2) collect stationary and controlled-motion S3 recordings while comparing raw acquisition and stored samples; (3) define pulse-quality acceptance criteria and rerun timing/morphology evaluation with all difficult events retained.

End with a 3–5 sentence recommendation to the engineering team: what is presently known, what is uncertain, what should not be reported as a valid pulse measurement, and the next experiment. You are free to make any valid assumptions on the hardware sensors, platforms and other sub-systems, as long as you clearly declare them in your response.

Recommendation to the engineering team: The current dataset shows a one-second packet gap in S5 and substantial high-frequency variation in S3, but neither observation alone identifies the responsible physical fault. Incomplete intervals and low-quality events should not be reported as reliable pulse measurements without an explicit quality decision. I recommend first tracing S5 packet loss through sender/receiver logs, then recording S3 under controlled stationary and movement conditions while comparing raw and stored samples. The 8 Hz filter reduced rapid variation and retained most detected peaks in this exploratory test, but morphology and boundary behavior require more validation before claiming improved accuracy.

4. Two Short Judgment Situations (10 minutes total, with disclosure)

A. Before a demonstration, someone asks you to hide a low-quality flag because it makes the display look less impressive. What would you do first, and how would you explain your recommendation to the team? (2–3 sentences)

Response A: I would first verify why the quality flag was raised and whether the recording meets predefined quality criteria. I would not hide the flag merely to improve the demonstration; I would explain the uncertainty, document the finding, and show the measurement as requiring review if the quality issue remains.

B. You announce that a filter improves timing; your mentor points out that difficult samples may have been removed from the comparison. What check would you run, and how would you update your conclusion if the mentor is correct? (2–3 sentences)

Response B: I would identify which samples were excluded and rerun both algorithms on the same representative dataset using consistent inclusion criteria, reporting exclusions and unmatched events. If the improvement disappears when difficult samples are included, I would revise the claim and report the result as limited to the evaluated subset rather than generalizing it.

5. Submission and Tools

Submit: (i) your notebook, pseudo-codes and python or processing scripts, (ii) all sketches, plots and illustrations, (iii) partial work that explains what failed and what you would do next is welcome.

You may use AI, documentation, tutorials or online sources. Please identify what tools you used, where they helped, what you checked independently, and which claim you trust least. We will ask you to explain and adapt one part of your work in the interview. AI use itself is not a negative criterion.

AI/tools disclosure: I used ChatGPT to help structure the analysis and draft/debug Python code for packet-integrity checks, plotting, PSD comparison, filtering, and peak-timing evaluation. I ran the code in Google Colab and reviewed the outputs. I independently checked the packet counts, missing sample interval, plots, and printed measurements. The claim I trust least is the physical interpretation of S3's high-frequency component: the available data show a difference, but do not establish its source. The 8 Hz filter is exploratory and has not been clinically validated.